AFFERENT-OMNI

ISO/TS 15066 & IEC 61508 Safety Compliance Audit Report

NASA/ESA-Grade Somatosensorial Robotics Control & Telemetry System

Physical AI Track - Nebius x NVIDIA Global AI Hackathon

Session ID: SESS-001
Target Object: Glass
Report Generated: 2026-10-28 00:00:18 UTC
Standards Evaluated: ISO 10218-1, ISO/TS 15066, IEC 61508, Sim2Real, TactoEval
Cognitive Copilot: NVIDIA Nemotron 3 Super 120B (Nebius Token Factory)

CONFORMITY STATUS: PASSED / CERTIFIED (5/5 - 100.0%)

Executive Summary	
Safety Shield Status	ACTIVE (Sim2Real Reflex Loop @ 50 Hz)
Target Object	Glass
ISO Verification Result	PASSED / CERTIFIED
Cognitive Copilot	NVIDIA Nemotron 3 Super 120B (Nebius Token Factory)

Standard	Check	Status	Value
ISO/TS 15066	Max force <= 2.0 N (collaborative)	PASS	1.8 N
ISO 10218-1	Safety margin >= 10%	PASS	97.6%
IEC 61508	Edge reflex latency <= 15 ms	PASS	8.8 ms
Sim2Real	Uncertainty <= 25%	PASS	9.0%
TactoEval	Damage prevention confidence >= 95%	PASS	99.2%

1. Tribological Material Profile

Material Class	soda-lime-glass
Young's Modulus (E)	70.0 GPa
Static Friction (mu)	0.22
Shear Limit	1.8 N
Slip Threshold	0.242 N
Safety Margin	97.6%
Surface Roughness (Ra)	0.05 um

2. Force Budget

Max Allowed Normal Force: 1.8 N

Recommended Grasp Force: 1.08 N

Minimum Slippage Threshold: 0.682 N

Force Budget Margin: 97.6%

3. Neuromorphic Reflex (Edge)

Edge Latency: 8.8 ms

Slip Detected: False

Reflex Override Engaged: False

4. Acoustic-Slip Spectrogram

Dominant Frequency: 314.3 Hz

Stick-Slip Energy: 1.084

5. Multi-Digit Topology

Active Digits: thumb, index, middle

Torque Balance Index: 0.857

Center of Pressure: [0.591, 0.642]

6. NVIDIA Nemotron AI Diagnosis

The somatosensory contact matrix confirms stable multi-digit engagement. No shear stress concentration zones exceeded the plastic